

Emergent parallels between language models and the brain

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How does the brain give rise to human capabilities like language processing or logical reasoning? This question has captured the minds of many cognitive scientists and neuroscientists, and indeed has a history dating back at least as far as the philosophers of ancient Greece. Until recently, humans were the only systems that we knew of that showed these capacities across broad domains. But after recent developments in the field of artificial intelligence, language models have begun to show some surprising parallels to complex cognitive capacities and the neural representations that underpin them—despite the architectures and learning paradigms of these models differing dramatically from biological systems. In this Primer I will provide a brief overview of language models, their construction and training, and the phenomena of natural intelligence that they might help us to understand. First, let us begin with a bit of history.

Connectionism: artificial neural networks as cognitive models

Forty years ago, a collection of cognitive scientists published the *Parallel Distributed Processing* books (Rumelhart et al., 1986), articulating an emerging perspective on the nature of cognition. The authors argued that cognition could be modeled through artificial neural networks, in which many small neuron-like units did simple computations in at the same time (i.e., parallel-in-time processing, distributed across many neurons), and in which learning occurred through updating the strength of the connections between these neurons. Thus, they followed earlier researchers in describing themselves as “connectionists” (although they emphasized their difference from earlier connectionists in their focus on parallel and distributed representations).

The connectionist perspective was in strong contrast to the dominant cognitive paradigm of the era, which viewed cognition as a series of logical inferences over symbolic propositions (e.g. Newell, 1980). By contrast, connectionists argued that their models could explain even high-level cognitive abilities—such as learning of the syntax and semantics of language, or procedures for logical reasoning—that were usually assumed to involve these kinds of logical operations. Furthermore, the connectionists argued that their models could capture the “microstructure” of cognition—making more detailed predictions about learning and behavior, and positing more concrete computational units that could potentially be more directly mapped onto real neural activity.

However, these ideas were dismissed by many cognitive scientists, who argued that these neural network models were fundamentally incapable of achieving human-like cognition (Fodor and Pylyshyn, 1988)—unless they did so by implementing a symbolic logic system, in which case, what was the point?

Despite (and in response to) these criticisms, the connectionists continued their research in modeling cognition with neural networks. These explorations included forays into language processing. In particular, Elman (1991) showed that neural network models that were trained to *predict the next word in a sentence*, using the earlier words as context, learned generalizable representations that captured aspects of the grammar and meaning of those sentences—for example, representing grammatical dependencies in consistent ways across different sentences.

Of course, the idea that learning through prediction could explain aspects of behavior and the computations underlying it was not new. It was already known to be a foundation of perceptual learning, and a growing literature was showing how simple behaviors—like the associations Pavlov’s

dogs had between a bell and their food—could be explained through predictive learning. Perhaps because of these behaviorist connotations, however, the idea that predictive learning could also explain “higher” forms of cognition (like language) was harshly dismissed by many cognitive scientists. Nevertheless, the idea of learning the structure and meaning of language by predicting the next word lived on, and was eventually adopted in machine learning.

The evolution of language modeling: Attention & scale

As computers became more powerful (your phone is probably around 1,000 times as powerful as the fastest supercomputers at the time the early connectionist works were published), machine learning researchers began to build larger models, and train them with larger quantities of data. Many of these researchers focused on areas such as vision or game-playing, but some continued to train predictive language models. The modern revolution in language modeling came primarily from two major changes in the past decade: a new neural network architecture, and dramatically increased scale of the models and their training data.

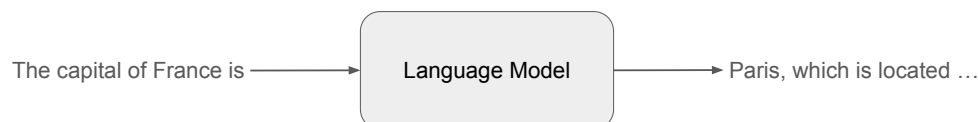
The architectural innovation (Fig. 1) is called a Transformer (Vaswani et al., 2017). Prior “recurrent” models effectively required a single representation (the present “state”) to capture every piece of information from the earlier text that might be useful in the future. By contrast, the Transformer architecture allows the language model to more flexibly *retrieve* information from earlier text as it makes each prediction. It does so through an operation that is evocatively described as “attention,” because it allows the model to *query* all of its representations from earlier parts of the text for relevant information, and thereby to selectively “attend” to particular parts of the context that are relevant to its current predictions. For example, In trained language models, some of these attention operations have been shown to follow grammatical structures (e.g., attending from a verb to its subject) as well as content ones (e.g., attending from a mention of someone’s name later in a story to their earlier appearances within it). Others appear to follow more abstract structures—such as repetition or parallelism—or more difficult to interpret, distributed patterns across the text. This flexible ability to learn to attend to different patterns within text is essential to the generality of modern language models.

(Of course, many other architectural features of modern language models—such as the use of normalization or sparsity—might seem familiar from neuroscience; these parallels are rich and interesting, but beyond the present scope.)

The second feature that differentiates modern language models is scale: both the scale of the models, and the scale of their training. Let’s talk about the model scale first. The first connectionist language models had a few layers of neurons, with a total of (roughly) 10,000 individual parameters—i.e., connection strengths between neurons—that could be learned. Modern language models might have **hundreds** of transformer layers, and **trillions** of parameters that can be learned. It’s hard to know what’s a reasonable scale comparison, but one might loosely guess that the number of parameters in these models is only a couple orders of magnitude smaller than the number of synapses in a human brain. (In terms of per-neuron synapses, LMs often have something in the range of 4,000–32,000 input synapses to each neuron for published architectures, which is only mildly larger than the range one might expect for a cortical pyramidal neuron.)

With all these parameters, the models can learn from a great deal of data. Language models are typically trained on large corpora of text, usually from the internet. In fact, models learn from tremendously more language than any human does (Frank, 2023); language models are often trained on trillions of words of data, while an average human optimistically hears a few hundred million words before adulthood. However, human experience is not solely in language. We receive much more experience in other modalities. Nevertheless, these differences in scales (and types) of experience make it more complicated to draw inferences about natural intelligence from studying language models.

Language models predict text continuations



... by predicting one word at a time, using layers of transformer blocks that integrate earlier context

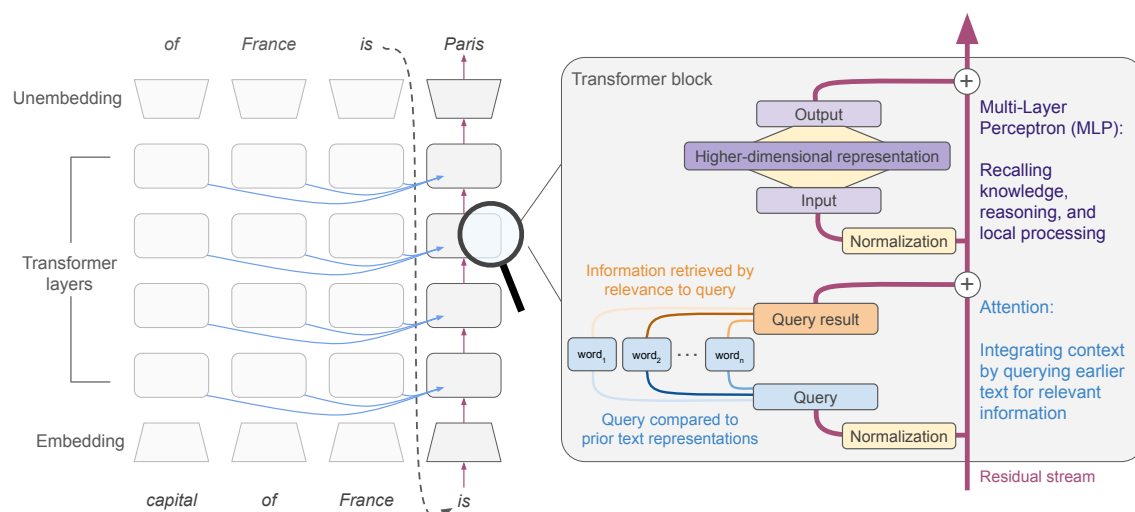
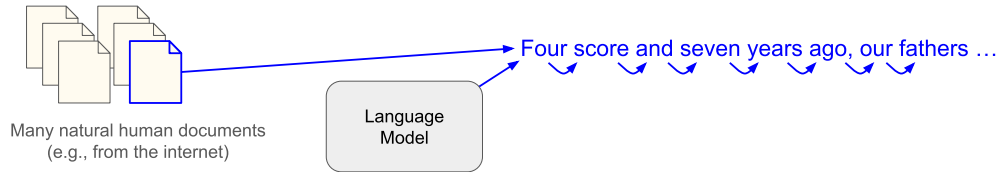


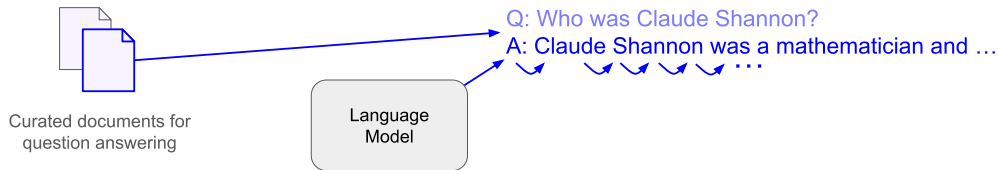
Figure 1: Language models predict how text will continue, one word a time.

(top) The model takes as input the beginning of some text (e.g., “The capital of France”) and predicts what word will come next (e.g. “is”). After it predicts the next word, its prediction (“is”) is provided back as input for predicting the word to follow (e.g., “Paris”). (bottom) To make these predictions, the model first embeds each input word into a high-dimensional space. This high-dimensional vector of neural activity forms the beginning of what is called the residual stream. The model then applies layers of transformer blocks to this residual stream. Each of these blocks consists of an attention operation, followed by a multi-layer perceptron. The attention operation integrates information from earlier words by allowing the model to produce a query which is compared to representations at early positions. The information values from earlier positions are then passed forward to the present one, weighted by how relevant that information is to the query. In this way, the attention operation allows the model to select relevant pieces of information and incorporate them into the current predictions. The results of this attention operation get added back to the residual stream. Next, the residual stream value gets processed by a multi-layer perceptron (or MLP) — a neural network that projects information to a higher-dimensional space and then back down. This allows the model to integrate the information from the words with its factual knowledge from other documents, as well as to identify what kind of information to query for next, etc. Again, the result of this gets added to the residual stream. After these transformer layers, the residual stream representation is compared to learned representations for all words. This comparison produces a probability distribution over next words, from which the actual next word is sampled.

1. Pretraining: learning to predict human text, one word at a time



2. Supervised fine-tuning: predicting answer text one word at a time, for specific tasks (e.g., following instructions and answering questions)



3. Reinforcement Learning (RL): given a question *without* a pre-written answer, the model writes an answer, and is rewarded for correctness, estimates of humans preferring the answer, etc.

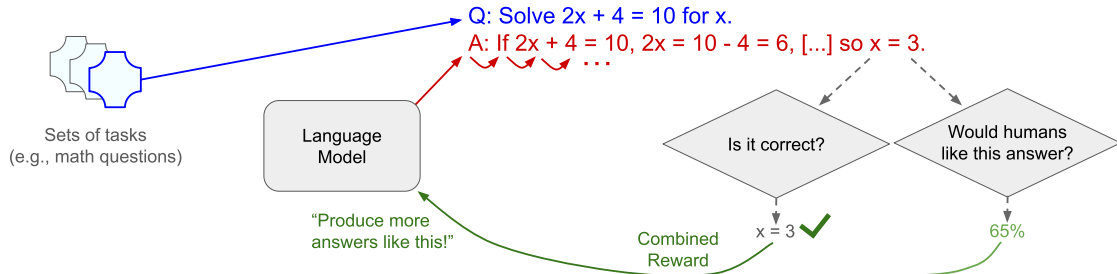


Figure 2: Language model training progresses through several stages.

In the first two stages of training, the model is trained to predict text that humans have written previously. In the first stage (pretraining), this text consists of a wide range of documents, to teach the model general language processing skills and knowledge. In the second stage (supervised fine-tuning), a smaller set of documents are similarly used to teach the model to follow instructions, answer questions, etc. In the final stage, reinforcement learning (RL), the model is given questions without a pre-written answer. The model then produces an answer, which is scored according to whether it is correct, as well as other features like whether a human would like this answer (relative to other answers to the same question). These scores get combined into a single reward, which is used as feedback to train the model. If the reward is high, the model will be trained to produce answers more like this in the future; if the reward is low, it will be discouraged from producing similar answers in the future.

Training a modern language model: from passive prediction to reinforcement learning

Given all this data, how does one train a language model? The typical process proceeds in a few stages (Fig. 2). The first stage uses a deceptively simple task: at each position in a document, the model is just trained to predict what word will come next. Specifically, the model produces a probability distribution over all the possible words that might come next. During training, the model is trained to put a higher probability on the word that actually *did* come next in the data. The training takes place by adjusting the connection strengths between the neurons in the model in order to make the correct predictions more likely. If a stronger connection strength between two particular neurons would have made the correct prediction more likely, it is (slightly) increased; conversely, if a stronger connection strength would have made the *wrong* prediction more likely, it will be (slightly) reduced. (For those familiar with calculus, the adjustment of each connection is determined by the derivative of the probability assigned to the correct next word with respect to that connection’s strength.) By making gradual updates across large quantities of text data, the model gradually starts to make more and more accurate predictions about the word to come.

Through this predictive training process, the model effectively learns a probability distribution over how a document will continue—though this probability distribution is implicit; it has to be evaluated one word at a time.

It turns out that this learned distribution can capture interesting and nontrivial features, ranging from the grammar of language to more complex abilities like being able to answer questions. For example, if you give a model a question, and let it generate—one word at a time—an answer that might continue the text, the answer will often be correct. Or, if you give it a few examples of working through arithmetic problems (e.g., “ $3 \times 12 = 3 \times 10 + 3 \times 2 = 30 + 6 = 36$ ”), it will sometimes be able to work through a few steps to answer another problem. However, as a probabilistic model it can also produce terrible answers or pure nonsense some of the time. (Indeed, terrible answers often might be consistent with internet text on which it was trained.)

In order to make this model more reliably useful, a second stage of “supervised fine-tuning” training is applied. This involves focused training on answering questions correctly, as well as other properties that the model developers would like the models to have (e.g., *not* answering questions like “how can I rob a bank and get away with it”).

The third stage moves beyond basic prediction of text to *reinforcement learning* (RL). In reinforcement learning, the model is given a context (such as a question) and is then allowed to write out its intermediate “reasoning trace” in natural language until it decides it is ready and generates a final answer. This reasoning trace and final answer are generated one word at a time, just as the model learned to predict language above. Importantly, the ability to generate this reasoning trace relies on behaviors the model has seen in training—for example, the way that humans reason in an essay—but the model can apply these abilities to answer new problems.

Once the model produces its final answer, it is rewarded according to how good the answer was by various measures. For example, if the model was asked a math problem, it might be rewarded for correct answers and penalized for incorrect ones. Or, the reward might be based on another model’s estimate of how easy-to-understand or friendly the answer is, or on whether it might be harmful to the user, etc. Rewards can be used to specify various desired behaviors.

The reward that the model received is used to update the way that the model produces intermediate reasoning tokens and the way it produces the final answer. For example, if the model produced an answer which gets high reward (e.g., a correct answer to a math question), its parameters are updated so that it will produce similar thinking processes and answers for similar questions in the future. If the model gets the answer wrong and is penalized, it will be updated to produce *less* similar thinking tokens and answers in the future. Thus, by using rewards the model can be trained to do things—such as writing complex programs or solving math problems—even if the creators do not *know* exactly the right way for the model to approach those problems. (This possibility is

in contrast to the earlier, supervised fine-tuning stage, in which the model creators must specify exactly how the model should produce the answer.)

After these final stages of training, the model resembles the chatbots you might have interacted with online—which generally have some ability to answer questions, perform tasks they are given, etc. The earlier, purely-predictive model is often referred to as a “pre-trained” model, and the final model further trained with supervised finetuning and RL is often referred to as a “post-trained” model.

So, what might these models have to do with natural intelligence?

Models trained on natural data predict neural activity

In the early 2010s, neuroscientists became excited about AI *vision* models that were trained to recognize objects in images. These models turned out to produce patterns of internal activity that were highly predictive of *biological* neural activity in visual cortex, including in higher, semantic regions (Yamins et al., 2014)—linear regressions from the *models’* internal activity could predict brain activity on new images surprisingly well. These findings were exciting because, although neuroscientists had general theories about the high-level function of these regions, prior models could not predict details of neural activity well.

It was especially surprising that AI vision models could predict neural activity, given that their architecture and learning experiences are so different than biological systems. These findings stirred excitement that there might be some common computational principles that would be shared across a wide range of systems solving the same problems (such as object recognition)—and thus that studying the computational principles at work in AI systems might help to understand the organization and function of the brain (Kanwisher et al., 2023).

Thus, as language models began to show increasing competence in predicting language, researchers naturally asked whether their internal activity could predict neural activity in language-processing regions of the brain. Indeed, in 2021 researchers found that the language models available at the time could predict much of the activity in the brain when predicting language (Schrimpf et al., 2021). In fact, linear regressions from the internal activity of some language models were able to predict an individual’s brain activity equally as well as activity from all the remaining humans could. This predictivity emerges early in language model training—from a human-like quantity of language (Hosseini et al., 2024), rather than the inhuman scale of training that modern assistants receive—and only requires the next word prediction phase of training (rather than RL or later stages). Furthermore, these model predictions are not simply correlational; they enable researchers to actually activate or suppress activity in language regions of the human brain, by using the model predictions to optimize the sentences to present (Tuckute et al., 2024).

The data from which language models learn is extremely different from the way that humans learn language. Humans learn from audition rather than text, in active communication rather than passive absorption, and so on. Thus, these surprising parallels suggest that predictive learning of high-level language—without the details of auditory cortex, or other specialized, language-specific brain-circuitry—may in some sense be sufficient to capture much of neural activity. Prediction is a simple task, but learning to predict gives rise to rich structure.

Behavioral parallels to cognition, in language and beyond

Furthermore, language models do not only predict neural activity. These models also reflect aspects of human cognition, from lower levels of linguistic processing to higher-level reasoning. For example, consider the most basic task used for training these models: predicting the next word. It turns out that the probability that the models assign to a word predicts how surprised humans will be when reading that word (Wilcox et al., 2020). These models can also learn more complex structures that had previously been argued to be impossible to learn without innate knowledge of grammar (Wilcox

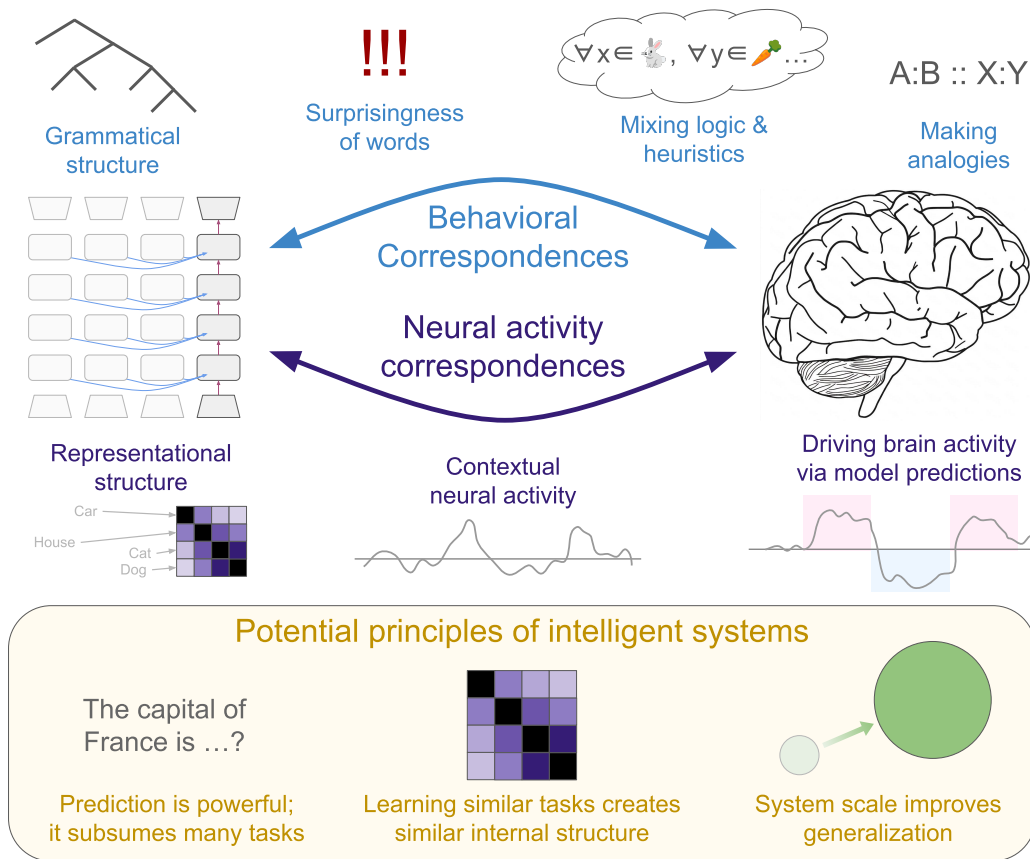


Figure 3: Language models show behavioral and neural correspondences to the brain, which hint at possible principles of intelligent systems.

Behavioral correspondences range from grammatical structure to patterns of reasoning. Neural correspondences occur at the levels of representational structure and contextual activity; the correspondences have sufficient fidelity to allow driving brain activity by using model predictions to optimize stimuli, for example. These behavioral and neural parallels suggest potential principles that may be common across many intelligent systems: the power of prediction, the constraints on internal structure imposed by learning similar tasks, and the importance of system scale for generalization.

et al., 2024). These results pose challenges to the dominant linguistic theories of the past half-century (Piantadosi, 2023; Futrell and Mahowald, 2026). The success of these models instead favors accounts of language that have emphasized the role of general-purpose learning systems applied to language inputs.

Surprisingly, these models also appear to reflect some cognitive capacities beyond language. For example, consider reasoning about logical problems. Human reasoning on these problems does not perfectly follow the rules of formal logic, but instead depends on the content that we are reasoning about in sometimes illogical ways—for example, often accepting an argument when they believe the premises and conclusion, even if the argument’s logic is flawed—and language models mix logic and content in similar ways (Lampinen et al., 2024). Likewise, the ability to reason by analogy is considered a key component of human cognition, and large language models show aspects of analogical reasoning (Webb et al., 2023). These human-like capabilities of models are often called *emergent*, because they emerge after training even though they are not the language-prediction task that models are directly trained to perform.

Even cases where language models *fail* to generalize may be informative about cognition. For example, language models show relational generalization deficits—such as failing to generalize from learning that “Tom’s mother is Mary” to “Who is Mary’s son?” These deficits can be improved by prompting the model to create additional training data by rewriting the original data in augmented forms. The relational generalization deficits seem to parallel some deficits shown by humans or animals with hippocampal lesions, and the remedy of creating new data by augmenting older data reflects some aspects of hippocampal-cortical interactions (Bonnen and Lampinen, 2026). Understanding patterns of generalization of language models under different data regimes can provide a useful analogy for the constraints and adaptations of natural intelligence.

Thus, language models are opening a variety of new paths to understanding various domains of neural computation and cognition.

Challenges of comparison

Despite the parallels described above, using language models to understand natural intelligence comes with theoretical and methodological challenges (Ivanova, 2025). A major problem is test “contamination”—because the models have been trained on large quantities of data, it is likely that they have seen descriptions of standard cognitive tasks in training, or even data human participants produced on those tasks. Thus, parallels to human cognition may simply be due to memorization, if the researchers are not careful. Many of the studies referenced above create entirely new experimental stimuli, to avoid contamination concerns. Moreover, memorization is an instance of a more general challenge: two systems behaving similarly does not necessarily imply that they arrive at that behavior via the same mechanisms.

There are likewise challenges with interpreting results that show high neural predictivity. High neural predictivity does not necessarily imply the systems are doing the same computation or learning in the same way. For example, the fact that models trained by next-word prediction exhibit high neural predictivity could be interpreted as evidence for predictive learning shaping the brain’s neural activity, but it could also be explained simply by the models and the brain both picking up on shared linguistic features through different mechanisms (Antonello and Huth, 2024). Moreover, we do not always know if we are correctly measuring which features of neural activity are “important” to predict; even weak neural signals can be important, but common measures of similarity favor stronger signals (Lampinen et al., 2026). We also probably overestimate how noisy brain computations are, by not properly accounting for the many features like cross-trial dynamics or attention that modulate neural activity. Thus, activity we dismiss as noise may nevertheless be a meaningful, important computation that our models fail to capture. At multiple levels, we do not really know how to be certain whether two systems are meaningfully doing the same thing in some general sense.

Finally, even where we are confident that language models’ computations meaningfully reflect

those in the brain, this parallel is not in itself explanatory. We have to go deeper to ask “why” these parallels emerge (Kanwisher et al., 2023); for example, experimenting with what they tell us about the task the brain is trying to perform, or how its adaptations may be driven by the structure of natural data. In some areas, research on language models is directly driving new explanatory progress. However, in other cases, language model research may be more useful for what it suggests about higher-level principles of learning and generalization, rather than for direct modeling of behavior or neural activity.

Possible lessons from language models: prediction, tasks, and scale

What broader lessons might we learn from language models? I will highlight three here (Fig. 3, bottom), which I believe might prove to be more general principles of the computational basis of intelligence.

The first is the power of prediction. Even language models pretrained solely on next-word prediction (before RL) show rich behaviors and reasoning over their inputs, including the ability to parallel certain aspects of human logical or analogical reasoning, or to learn and apply a novel function or concept from examples in context. This shows how simple prediction—when applied to sufficiently complex data—subsumes many tasks, and thus can give rise to complex structure and even adaptability.

A second lesson is that sufficient learning from the many tasks that prediction subsumes gives rise to similar internal structure, even in systems that are structurally dissimilar—and the larger and more diverse the set of tasks, the stronger the constraints on the structure that is learned (Cao and Yamins, 2024). That is, if two systems are learning roughly the same large set of difficult tasks—such as processing all of natural language—the two may simply be forced towards similar internal solutions. Language models and language circuitry in the brain could hardly be more different architecturally, and the formats in which they receive language barely resemble one another, yet their internal representations can be regressed onto one another relatively reliably. Thus, in a sufficiently powerful learning system, the tasks being solved may determine most of how the system functions.

A corollary lesson is the importance of the rich diversity of real-world data. Language models show the most general intelligence of any computational system to date, and a key contributor is the natural data they are trained on. Controlled studies show that more varied data leads to better generalization on truly novel tests (Misra and Mahowald, 2024). Thus, the variety and richness of natural experiences may be an important contributor to the generality of intelligence.

The final lesson is the importance of scale. A fundamental property of language models is that they exhibit a variety of *scaling laws*: predictable mathematical relationships between features like the size of the model or the quantity of data it has been trained on, and how accurately it predicts language. One important law is that (all else being equal), larger models generalize *better* than smaller models from the same training data. Thus, the scale of a system can be an important feature in how well a system performs in new situations. Perhaps for brains as well, scale is an important feature (cf. Herculano-Houzel, 2012).

From these last two points, we can conclude that overuse of small scale models applied to simplified data—the modeling paradigm often used in cognitive science and neuroscience—may mislead us about the principles of natural intelligence. Indeed, we can see instances of this where the empirical successes of language models have refuted claims about the impossibility of learning various language features, as noted above. Thus, the study of natural intelligence may need a paradigm shift to allow accommodating larger-scale models and the richness of natural experience in our theory building (Carvalho and Lampinen, 2025).

Conclusions

Language models are an exciting demonstration for cognitive science and neuroscience, and they open new directions for research. We are beginning to see how complex behaviors that begin to resemble natural intelligence can emerge predictably from a large-scale system doing a simple prediction task over large-scale natural data. We are beginning to realize some of the vision of the connectionists—showing how systems that learn by gradually updating the connections between neurons can come to exhibit some of the complex features and broad abilities of natural intelligence. Perhaps by understanding these phenomena, we can understand some of the general principles underlying intelligent systems.

Declaration of Interests

The author is employed by Anthropic.

Further reading

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